

PROJECT INTERMIX

Trust + Speed

v1.4.1

Portable architecture and operator handoff

Pixel 10 Pro / Tensor G5 / Android 17 / Termux
Gemma 4 E4B-it / LiteRT-LM 0.16.1 / fixed 8K context
SQLite FTS5 / Textual / optional Resonance

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Portable architecture and operator handoff

Release date: 2026-08-25 Primary target: Google Pixel 10 Pro, Tensor G5, Android 17, Termux Local inference: Gemma 4 E4B-it through LiteRT-LM 0.16.1 GPU/OpenCL Physical context: fixed 8,000-token KV cache Primary database: SQLite with FTS5

1. Executive summary

Project Intermix is a local-first companion and bounded workspace agent designed for sustained use on a Pixel-class Android device. It combines a resident local language model, a cyan-purple Textual cockpit, selective SQLite continuity, controller-owned factual guards, a fixed workspace executor, optional web grounding, and an offline-safe Kokoro audio companion.

Trust + Speed v1.4.1 fixes a concrete failure mode: broad search could mistake a GitHub compatibility mention for the latest Python release. The controller now classifies exact claims, routes supported facts to official sources first, extracts a verified value outside the model, and validates that the final answer preserves both value and citation. The exact lookup lane uses only 2,200 selected input tokens, while safe engine warm-up overlaps retrieval when cold memory headroom permits.

No credential is stored in SQLite, prompt context, workspace, report output, or release archive. Optional providers are configured through a standalone hidden input tool and loaded controller-side on the next launch.

2. Device and runtime envelope

Component	Measured or configured state
Device	Pixel 10 Pro, Tensor G5, Android 17
Physical RAM	About 15.2 GiB visible to Termux
Idle available memory	Typically 7.0-7.6 GiB without Resonance
Resident HOT available memory	Typically 2.6-3.7 GiB
Model	gemma-4-E4B-it.litertlm, about 3.5 GiB
LiteRT-LM	0.16.1 Python API and CLI
Backend	GPU/OpenCL; CPU fallback components where required
Model initialization	About 6.5-8 seconds in measured smoke tests
Warm first-text delay	About 1.2 seconds in the direct resident smoke test
Storage	Project data, compiled caches, and model under Termux app storage
Cooling	External Peltier cooling; observed operating target around 25 C

Android may terminate a process with Signal 9 under memory pressure. Intermix does not assume that a session reset will occur first. Every user turn, mission event, memory update, and verified task state is committed independently so a restart rehydrates from SQLite and the mission ledger.

3. Runtime topology

- Textual accepts a user prompt and immediately renders local UI state.
- The deterministic freshness and exact-claim classifiers select local, grounded, verified-lookup, creation, or workspace behavior.
- SQLite retrieval selects only relevant continuity inside the active token

ceiling. Exact lookup deliberately omits unrelated long-term blocks.

4. If grounding is needed, official or configured providers run under a strict request budget. A cold resident engine may warm concurrently when safe.
5. Prompt Builder assembles identity, mandatory contracts, selected evidence, current input, and only the context permitted by the active response mode.
6. Gemma generates language or bounded workspace tool proposals.
7. Hidden memory protocol, exact-number integrity, exact-fact validation, and citation guards filter the response before display and persistence.
8. The TUI renders final Markdown once. Optional Resonance can receive only a fully generated, filtered, committed response after explicit user action.

4. Exact-fact trust path

Claim contract

The controller distinguishes these claim types from ordinary topical search:

- latest stable version;
- latest prerelease version;
- release date;
- current price when a concrete subject can be identified.

Each exact contract carries subject, channel, reason, and a canonical cache key. Stable and prerelease facts cannot share a cache entry.

Entailment gate

Topical overlap is not proof. Evidence must assert the requested relationship. For example, "package X supports Python 3.15" is a compatibility statement and cannot prove that Python 3.15 is the latest stable Python release.

Official resolver

Python version questions use Python.org Downloads first. The controller parses official version candidates, separates stable from alpha/beta/RC releases, selects the correct maximum, and emits a structured fact with value, channel, source, provider, authority, and retrieval time.

Answer guard

The model draft remains hidden until it contains the exact verified value and a valid evidence citation or official URL. A conflicting version causes rejection. One small repair epoch is allowed. If repair still fails, the controller emits a deterministic official-source answer rather than exposing speculation.

5. Physical context policy

Mode	Input ceiling	Output reserve	Primary purpose
Verified lookup	2,200	5,800	Exact current fact and brief context
Brief	7,200	800	Compact answer
Normal	6,800	1,200	Ordinary conversation, about ten sentences

Mode	Input ceiling	Output reserve	Primary purpose
Deep	5,800	2,200	Structured explanation
Workspace	5,200	2,800	Tool actions, diagnostics, repair
Create	4,000	4,000	Video scripts, lore, long-form artifacts

Every row totals the fixed 8,000-token cache. The lookup lane does not increase KV memory. It reduces prefill by excluding session checkpoint, second-brain events, semantic recall, and memory-write protocol unless the request needs those systems.

6. Adaptive resident-engine policy

Profile	Available memory	Effective idle retention
Performance	At least 2,560 MiB	600 seconds
Balanced	2,048-2,559 MiB	90 seconds
Pressure	1,280-2,047 MiB	20 seconds
Critical	Below 1,280 MiB	5 seconds

Cold prewarm requires at least 5,632 MiB available. A loaded engine may be reused without reapplying the cold threshold. These defaults fit the reported Pixel measurements but remain environment-configurable. The controller treats warm-up as an optimization; any warm-up failure leaves normal backend fallback intact.

7. Web grounding and provider policy

Hard limits remain six HTTP requests, three concurrent requests, eight accepted results, URL deduplication, relevance/freshness gates, and provider cooldowns.

Route	Role
Python.org	Exact official Python release facts
SerpAPI	Optional structured Google result fallback
TinyFish Search	Optional structured lawful search fallback
Brave, Tavily, Exa, SearXNG	Optional configured search wave
GitHub REST	Explicit repository, issue, commit, and project-tech queries
PyPI, Crossref, arXiv	Query-specific registry and research adapters
Google News, Bing News, Wikipedia	Keyless discovery under policy gates
DuckDuckGo HTML/Lite	Sequential final fallback with one shared breaker

Intermix does not rotate IP addresses, evade CAPTCHAs, or bypass provider terms. When evidence is inadequate, the correct result is a visible Grounding Guard.

8. Provider Vault

Run `intermix-providers` outside the cockpit. Secret prompts are hidden. The tool atomically writes `~/.config/intermix/providers.env` with mode 600 and only allowlisted fields. The controller loads this file on startup without replacing an explicitly supplied environment variable.

The cockpit exposes readiness and cooldown state through `/providers status` and `/web status`; it never displays values. `/providers setup` explains how to leave the cockpit and open the vault. A credential that has

appeared in any chat or screenshot must be rotated before use.

9. Memory and second-brain boundaries

SQLite retains immutable conversation messages, session checkpoints, durable memories, typed timeline events, mission records, fact watches, prompt reports, and provenance. FTS5 supports lightweight local retrieval without a separate embedding model.

Wellbeing data is eligible only when the user explicitly states it in the first person. It is query-gated, retention-limited, forgettable, purgeable, and omitted from reports by default. Intermix does not infer or store a diagnosis. SQLite is not encrypted by Intermix; Android app-private storage is the primary boundary.

Virtual context is not infinite attention. Raw history can remain durable while each turn selects a bounded set of facts, events, recent dialogue, verified work, and active mission state. Exact lookup uses the smallest safe selection.

10. Workspace autonomy

Automatic read, write, and test actions are confined to the fixed Sovereign workspace. Tool tags execute in emitted order. A failed action stops later actions in that epoch. A following epoch receives verified feedback and may repair. Meaningful writes require a post-write test before mission completion.

Deletion is never automatic: it becomes a review request with hash revalidation and recoverable handling. Reserved control state cannot be modified by the model. Network findings are untrusted data, not executable instructions.

11. Interface behavior

The Textual cockpit preserves the established cyan-purple visual system with white/grey response text. Assistant streaming uses an inexpensive plain text widget and performs one Markdown render at completion. Long diagnostics remain foldable. The new verified-fact card uses cyan for trust state, white for the value, purple for source identity, and grey for evidence metadata.

Hybrid fact mode shows the card plus brief Core context. Core mode shows only the guarded answer. The setting persists in SQLite and can be toggled locally; it is not delegated to the model.

12. Resonance compatibility

Resonance v1.3.1 remains optional and offline-safe. The Debian Kokoro cockpit, model, voices, and WAV archive are not changed by this release. Termux owns Android playback after Debian renders a verified WAV. Only a fully generated, protocol-filtered, committed response can enable VOICE, and the user remains the final consent boundary.

13. Installation, rollback, and recovery

The installer refuses to run while Sovereign is active, runs all tests, verifies Textual/LiteRT/FTS5 and storage headroom, creates a bounded `pre_trust_speed_install_*` snapshot, installs modules atomically, installs the Provider Vault command, compiles Python, verifies SQLite v2, and ends with a non-mutating legacy migration report.

It does not launch Sovereign, load Gemma, start Debian, enable voice, change WAV files, delete memory, or include provider credentials. If migration reports `importable: 0`, launch with `sovereign`. Preserve the printed rollback path until daily testing is stable.

14. Validation evidence

The portable build contains 55 tests. In the build environment, 53 executable tests pass and two Textual interaction tests are intentionally deferred to the Pixel installer, where Textual 8.2.8 is verified. Coverage includes:

- exact claim classification and stable/prerelease separation;
- rejection of compatibility mentions as latest-version proof;
- official Python resolver short-circuit;
- bad exact draft hiding and repair;
- 2,200-token lookup budgeting;
- Provider Vault permissions and secret-free status;
- SerpAPI and TinyFish response parsing;
- adaptive resident profiles and single engine warm-up;
- prior numeric, memory, privacy, grounding, workspace, and migration behavior.

15. Immediate Pixel acceptance checks

1. Install with Sovereign closed and preserve the rollback path.
2. Run `intermix-providers` outside the cockpit; configure only desired keys.
3. Launch `sovereign`; run `/providers status`, `/engine status`, and `/status`.
4. Set `/facts style hybrid`; ask "What is the latest Python version?"
5. Confirm Python.org fact card, identical Core value, citation [1], and a lookup prompt no larger than 2,200 tokens in `/inspect context`.
6. Toggle `/facts style toggle`; repeat and confirm the card disappears while the guarded value remains identical.
7. Compare cold and immediate warm latency; inspect residency profile and free memory after each turn.
8. Continue daily use and record provider quality, guard activations, irrelevant recall, memory pressure, storage growth, and any Signal 9 event.

16. Deferred work

The next production increment should add official resolvers only for recurring measured needs, such as Android security bulletins, LiteRT-LM releases, Textual, and package registries. TinyFish Fetch can be evaluated for bounded JavaScript page extraction after Search proves useful. Document Studio, richer PDF creation, and proposal-only self-audit remain separate releases.

The experimental AI social Observatory remains isolated and deferred. It must use its own database, memory, identity, quotas, and kill switch, with no access to personal memory, credentials, workspace tools, audio, or autonomous posting.